

RESISTIVE LOAD BANK PART 1 -

GENERAL

1.1 SUMMARY

- A. This Section includes a forced air-cooled resistive load bank. These units shall provide a manual exercising system for the engine generators.

The contractor shall supply two forced air-cooled resistive load banks with the following specifications and to remove the existing load banks from the premises

- Load Bank 1: Rated for 880 kW capacity.
- Load Bank 2: Rated for 265 kW capacity.

1.2 SUBMITTALS

- A. Product Data: Provide physical data of the load and control enclosures including dimensions, weights, cable access, mounting, operating clearances, and details of the component arrangements.
- B. Wiring Diagrams: Provide detailed wiring diagrams of the power and control circuits, including point-to-point wiring, wire numbers, connection point identification, components labeled by the device legend.
- C. Field Test Reports: Indicate and interpret test results for compliance with performance requirements.
- D. Product Test Reports: Certified copies of manufacturer's design and routine factory tests required by referenced standards.
- E. Sound-Level Test Reports: Certified copies of manufacturer's sound-level tests applicable to equipment for this Project.
- F. Maintenance Data: For equipment to include in maintenance manuals. Include the following:
1. Lists of spare parts and replacement components recommended for storage at Project site.

1.3 PERFORMANCE REQUIREMENTS

- A. The Load Bank shall be a complete system with all necessary pilot and power control wiring, and devices to provide a functional system. The only additional wiring shall be site required electrical wiring to the power source and interconnections from the control panel to the load enclosure.
- B. The load bank shall have the capability of load testing its corresponding engine generator up to its 100 percent rating, to verify its performance.
- C. Sound Levels shall conform to requirements in Chapter 12 of the OBO Building Code.

1.4 QUALITY ASSURANCE

- A. Comply with requirements of the OBO Electrical Code (NFPA 70, "National Electrical Code" as amended by OBO).
 - 1. Electrical Components, Devices, and Accessories: Listed and labeled as defined in NFPA 70, Article 100 by a testing agency acceptable to Project Director/COR based upon input from OBO/PDCS/DE/EE, and marked for intended use.

1.5 PROJECT CONDITIONS

- A. Product Selection for Restricted Space: Drawings indicate maximum dimensions for the load bank, including clearances. Comply with indicated maximum dimensions.
- B. The load banks shall be designed for continuous operation at full load and constructed using materials suitable for coastal environments
- C. Environmental Limitations: Rate equipment for continuous operation under the following conditions, unless otherwise indicated:
 - 1. Ambient Outdoor Temperature: 21.9 deg C to 32.4 deg C.
 - 2. Relative humidity: Up to 95 percent non-condensing.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

- A. Manufacturers: Subject to compliance with requirements, provide products by one of the following:
 - 1. Load Technology, Inc. (LOADTEC).
 - 2. Avtron Manufacturing, Inc.
 - 3. Alber Corp.
 - 4. Cannon Load Banks.
 - 5. Kohler Co.
 - 6. Post Glover Resistors, Inc.
 - 7. Power Conversion Systems.
 - 8. SBP Industries.

2.2 CONFIGURATION

- A. The load bank shall consist of the following items:
 - 1. The load bank enclosure containing the load resistors, cooling fans, and power control devices.
 - 2. A control panel containing the required pilot controls and indicators.
- B. Refer to the drawings for location of the load bank enclosure and control panel.

2.3 RATINGS

- A. Rating shall correspond to 100 percent of the generator KW output rating at 1.0 power factor.
- B. Load steps shall be rated at 25 KW, 50 KW, 100 KW or a combination of these steps.
- C. Duty cycle shall be continuous at rated ambient conditions.
- D. Tolerances:
 - 1. Total: plus or minus 5 percent of nominal rating.
 - 2. Balance: plus or minus 2 percent of mean phase.

2.4 ENCLOSURES

A. Load Enclosure:

- 1. The enclosure shall be suitable of outdoor installation with bolt down provisions for permanent mounting to a concrete pad.
- 2. The enclosure shall be provided with lifting eyes and captive fork lifting tubes.
- 3. The discharge air flow from the unit shall be vertical. The unit shall be constructed to prevent re-circulation of the cooling air.
- 4. The enclosure shall be constructed to isolate the heated air flow through the load resistor chamber, from the control section containing the power control and distribution components.
- 5. The control section shall be NEMA 4X stainless steel. rated with gasketed access doors. The load section shall be NEMA 3R stainless steel with fixed louvers and guards on the intake and exhaust openings.
- 6. The enclosure shall have adequate space to install the conduits from the power source. .
- 7. Exhaust louvers, if used, shall be fabricated of stainless steel.

B. Control Enclosure:

- 1. The control enclosure shall be indoor type, suitable for wall mounting.
- 2. Enclosure shall be NEMA 4 or 12 rated with hinged and gasketed access door.
- 3. All control components shall be sub-panel mounted with pilot controls and indicators mounted and wired to the exterior door.

C. Paint:

- 1. All metal surfaces shall be properly prepared, primed, and painted. There shall be no raw surfaces except for stainless steel or panels that have platings.
- 2. Surfaces to be painted shall be prepared in accordance with the Society for Protective Coatings (SSPC) Surface Preparation requirements:
 - a. Standard SP1.
- 3. All surfaces to be painted shall be primed using an etching primer intended for use on galvanized steel and applied in accordance with the manufacturer's recommendations.
- 4. All primed surfaces shall be finish coated using an exterior U.V. resistant polyurethane enamel paint applied in accordance with the manufacturer's recommendations.

2.5 LOAD RESISTORS

- A. The load resistors shall be open wire wound type that are continuously supported the length of the resistor structure. If the resistor alloy should fail, the resistor material shall be held captive to prevent contact with any surrounding resistors or ground.
- B. The resistor's continuous support shall consist of segmented ceramic tubes assembled on stainless steel rods with fastening provided to compensate for thermal expansion of the ceramic tubes.
- C. The resistors shall be fabricated of a corrosion resistant chromium alloy that has a minimum continuous operating temperature of 1000 deg C.
- D. The resistor shall not operate at any point in excess of 67 percent of the alloy's maximum continuous operating temperature at ambient extremes.
- E. The resistors shall be installed in multiple field removable trays that are small enough to facilitate easy maintenance. Each tray shall be individually removable without affecting any adjacent tray.
- F. The resistor terminals shall be brought to the exterior of the tray through high grade electrical ceramics. No plastic or polyester glass laminates shall be used in any part of the resistor assembly.
- G. The resistor terminations shall be formed from the alloy material or have fabricated terminals of the similar material welded to the alloy.
- H. The resistors shall not require a cool down period. Failure shutdown of the cooling fan during operation of the resistors shall not shorten their life expectancy.

2.6 COOLING SYSTEM

- A. The Load Bank shall be of the forced air cooled type.
- B. The cooling fan(s) shall be one piece cast aluminum and direct driven by a three phase, TEFC, induction motor.
- C. The motor shall be rated at the maximum brake horsepower of the fan propeller for the applied static pressure load.

2.7 PROTECTIVE SYSTEM

- A. Cooling Failure:
 - 1. A failure circuit shall be provided to monitor the cooling system operation and to disconnect the load steps before damage can occur.
 - 2. An airflow switch for each cooling fan shall be provided to monitor fan operation.
 - 3. A temperature switch at the load chamber exhaust of each cooling fan shall be provided to monitor excessive exhaust temperature.

B. Fuses:

1. Each individual load section/load step shall be fused to protect its wiring and control device.
2. Provide required fuses to protect the control power transformer and wiring.

C. Cooling Fan:

- D. Each cooling fan shall be protected by a thermal magnetic molded case circuit breaker in combination with a contactor and thermal overload relay.

2.8 WIRING AND LOAD BUS

A. Wiring:

1. The load wiring shall have a minimum rating of 150 deg C.
2. The control wiring shall have a minimum rating of 105 deg C.
3. The wire shall be sized at 125 percent of the maximum ampacity at rated ambient.
4. Control wiring shall have sleeve wire numbers that are identified on the wiring diagrams.
5. External interconnect wiring shall be terminated at barrier terminal blocks.

B. Load Bus:

1. A main bus shall run the length of the control section of the load enclosure so that the bus mounted fuses will require only short jumpers to the load contactors.
2. The load bus shall be tin or silver plated copper and shall be sized for the temperature rise at the full ampacity rating of the load. The load bus shall have a maximum temperature rise of 35 deg C.
3. Mechanical lugs shall be provided to connect the power source. The number and size of the lugs shall accept the number and size of the conductors shown on the drawings.
4. Mechanical lugs shall be provided to accept the required equipment grounding conductors shown on the drawings.

2.9 POWER CONTROLS

- A. The load shall be controlled by contactors capable of continuous and cycling operations.
- B. All control devices shall be clearly and permanently labeled.

2.10 PILOT CONTROLS

- A. Provide the required control circuitry needed to achieve the functions described for the control switches and indicators.
- B. All exterior control devices shall be permanently and clearly marked for function and operational settings. All control devices shall be clearly and permanently labeled on the interior of the enclosure.

- C. All control switches and indicators shall be industrial oil tight type if the control enclosure is not mounted inside an appropriately NEMA rated control compartment within the load bank.
- D. Control panel shall have the following control switches:
 - 1. The Control Mode Switch will select the operational mode of the system. The switch shall have three positions for the functions of Auto-Off-Manual. In the AUTO position, the load bank shall automatically drop the connected load steps, and be ready to respond to the requirements of the sequence of operation as defined in Division 26 Section "Double-Ended Low-Voltage Switchgear", paragraph 2.11, in the event of the utility power failure or other cause that requires the power source to service its normal loads. In the OFF position, the load bank shall be locked out and all circuits reset. In the MANUAL position (to be used only for testing of the generator in the presence of the operator), the load bank shall automatically drop the connected load steps in the event of the utility power failure or other cause that requires the power source to service its normal loads. In this case, the operator shall manually change the position of the Control Mode Switch from the MANUAL position to the AUTO position. The load bank shall then be ready to respond to the requirements of the sequence of operation as defined elsewhere.
 - 2. Fan Control/Cooling Failure Reset switch to control the cooling fan(s) and reset any cooling failure alarms. The switch shall be three positions for functions of Off-On- Reset (momentary).
 - 3. Master Load switch, which allows simultaneous control of preselected load steps.
 - 4. Load Step switches to control each load step.
- E. The control panel shall have the following indicators:
 - 1. Fan running indicator with green lens to annunciate if the fan(s) is running.
 - 2. Cooling failure indicator with red lens to annunciate a fault condition.
 - 3. Load Step indicators with blue lenses to annunciate the energization of each load step.

2.11 CONTROL POWER

- A. Control Circuits:
 - 1. Control power shall not exceed 240 volts.
- B. Fan Power:
 - 1. Fan power shall be 400 volt, 3 phase, 4 wire, [50] Hz. Coordinate number of required circuits and ratings with equipment supplier.

2.12 EXECUTION

2.13 REMOVAL & INSTALLATION

- A. Install the load enclosure and control panels in accordance with the manufacturer's drawings and instructions in the location shown on the contract drawings.

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- B. Provide and install power and control wiring in accordance with the manufacturer's drawings and instructions.
- C. Provide and install a switch, within sight of the load bank, to disconnect all power to the load bank during servicing.
- D. Remove the existing load banks from the premises

2.14 FINAL ACCEPTANCE

- A. The Load Bank system shall be adjusted and inspected in accordance with the manufacturer's initial operation instructions.
- B. The entire system shall be operated to demonstrate the function of each control panel and the accuracy of the applied load.
 - 1. Each failure sensor shall be operated to demonstrate the operation of the safety system.